GROUP MEETING · DATA STEWARDSHIP

Organizing Our Data

Layouts, formats, and data packages for papers, using our own six datasets

1
Organize
one lab layout

2
Describe
one study card

3
Package
common formats

4
Submit
a frozen, citable slice

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Fill one study card; every format reads it

study_card.yaml
~20 facts, filled once

data_dictionary.csv
column, unit, method, ontology ID



README.md
for people

miappe.tsv
plant experiments

croissant.json
ML + Responsible AI

DOI form (DataCite)
citation

Like a passport: give your details once, and every checkpoint reads the same page. Nobody types a fact twice.

The minimum core: 20 facts per dataset

1 Title
2 Short description
3 Steward + ORCID
4 Owner / PI
5 Funding + award no.
6 Dates (each session)
7 Location / lab room
8 Device model, serial, firmware
9 Capture protocol + settings
10 ID scheme
11 Species + genotype IDs (or none)
12 Variables → data dictionary
13 Ground truth + method
14 Software + version + git commit
15 File formats
16 License + access tier
17 Known limitations
18 Intended use
19 Possible biases
20 Sensitive info removed?

Record on the day

Nothing can recover these later:

- dates and times
- device + settings
- protocol
- ID scheme
- calibration / reference files
- ground truth + method
- capture_log.csv row per session

Four habits that make data trustworthy

Stable IDs

One ID per plot, plant, sample, object, episode that never changes, e.g. 2026_AEA_R03_C12. Same ID in file names, tables and metadata. Genotypes also get the public ID (GRIN, G2F).

Label every value

value_type = observed (sensor), derived (computed: script + git commit) or simulated (model + version). Raw data stays read-only.

Check before use

Schema, units, ranges, duplicate files, ID mismatches. Broken files go to 99_review/. A new site or season is flagged, not rejected.

Splits that don’t leak

Random rows leak across genotype, season, site, object. Split by group (leave-site-out, leave-object-out) and save the split as ID lists.

Same project, three common layouts

Our lab layout

```
maize-lidar-2025/
  README.md
  00_docs/
  01_metadata/
  02_raw/01_2025-07-14/
  03_processed/
  04_results/
  05_code/
  99_review/
```

Cookiecutter Data Science

```
maize-lidar-2025/
  README.md
  references/
  data/raw/
  data/interim/
  data/processed/
  src/  notebooks/
  models/
  reports/figures/
```

Hugging Face dataset

```
maize-lidar-2025/
  README.md  (dataset card)
  data/
    train-*.parquet
    test-*.parquet
  croissant.json (auto)
```

Mapping: 02_raw = data/raw · 03_processed = interim + processed · 04_results = reports + models · 05_code = src + notebooks

Packaging formats stack like nesting boxes

BagIt bag: manifest-sha256.txt

data/ = our lab layout

`croissant.json`
ML view + RAI

`ro-crate-metadata.json`
data + code + workflow

`datapackage.json`
table schemas

Format	Use it for
BagIt	big transfers, archives (checksums)
Croissant	any ML dataset; NeurIPS requires it
RO-Crate	data + code + workflow together
Frictionless	CSV tables that must validate
DataCite	the DOI / citation record

Open file formats, and what must travel with them

Data	Format	Must travel with it
UAV RGB / multispectral	GeoTIFF, JPG with EXIF	calibration panel, flight log, CRS
LiDAR / SfM points	LAS / LAZ	CRS (EPSG) in header, software version
NeRF / 3DGS	PLY + transforms.json	camera poses, settings, git commit
Hyperspectral	ENVI .hdr+.bil, Zarr	wavelengths, white/dark references
Robot episodes	LeRobotDataset	calibration files, fps, joint units
Tables	CSV, Parquet	data dictionary: units + ontology IDs
Genotypes	VCF (.vcf.gz)	reference genome version (B73 v5)
Vectors (plots)	GeoPackage, GeoJSON	CRS

Example: a UAV-LiDAR season, before and after

Before

```
LiDAR stuff/
  final_v3_REAL.laz
  cloud merged FINAL.las
  DJI_0001.JPG
  ...
  DJI_2400.JPG
  plot map FINAL.xlsx
  heights.csv
    (no units,
     plot IDs "3-12")
```

After

```
2025_AEA_maize-uav-lidar/
  README.md
  01_metadata/  study_card.yaml  ids.csv
                plot_map.gpkg  capture_log.csv
                data_dictionary.csv  croissant.json
  02_raw/01_2025-06-20/lidar/
         20250620_AEA_f01.laz      (read-only)
  03_processed/01_2025-06-20/
         20250620_AEA_lidar_merged_v1.laz
  04_results/plot_traits_v1.parquet
  05_code/height_extraction.py
  99_review/possible_duplicates/
```

Two more patterns worth copying

Sidecar per scan (hyperspectral ears)

Each scan carries its own settings file

```
02_raw/01_2026-09-10/
  E0001/
    E0001_view000.hdr
    E0001_view000.bil
    E0001_view000.json
      angle, exposure, lamp
  _refs/
    white_0910.hdr/.bil
    dark_0910.hdr/.bil
```

Splits as ID lists (multi-scale ML)

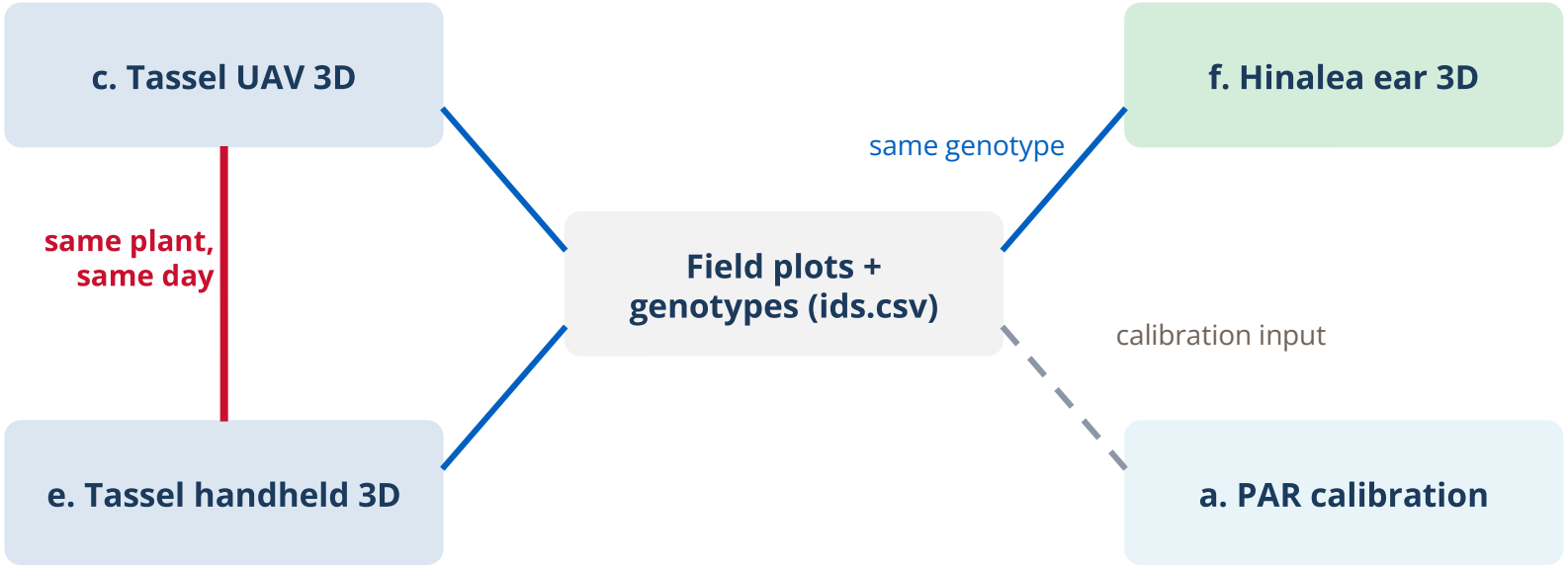
The rule is written down, not just done

```
splits/
  README.md
    no test genotype > 0.25
    kinship to training;
    leave-season-out
  train_ids.txt
  val_ids.txt
  test_ids.txt
```

Our six datasets: same layout, different extras

Dataset	MIAPPE	Keep as-is	Key extra facts
a. PAR sensor calibration	no	tidy time series	reference sensor + cert date, sky condition, sun angle, mounting
b. LeRobot, random objects	no	LeRobotDataset	robot calibration, fps, cameras, object list (size, mass, photo)
c. Tassel, UAV + 3D	yes	images, PLY, transforms.json	altitude, GSD, overlap, RTK + CRS, wind, growth stage
d. Sorghum stalk panorama	yes	images, panoramas	scale bar in frame, distance, stitching, caliper truth
e. Tassel, handheld + 3D	yes	images, PLY, transforms.json	camera settings, path, scale marker, in situ or cut
f. Hinalea ear, 3D	yes	ENVI/Zarr, spectral PLY	wavelengths, integration time, lamp, white/dark per session

Linked datasets are worth more than the sum



`links.csv:  source_id, target_id, link_type, rule, confidence`

For a paper: a frozen, citable slice

```
submission_NeurIPS2026_maize-lidar/
  README.md
  DATA_AVAILABILITY.md
  CITATION.cff     LICENSE
  data/            (only what the paper uses)
  metadata/
    croissant.json (core + RAI)
    miappe.tsv
    data_dictionary.csv
    provenance.json
    MANIFEST.sha256
  code/reproduce_fig3.py ...
  splits/train_ids.txt ...
  REVIEW_ACCESS.md (anonymous link)
```

Every figure maps to data + script + commit

Paper	Data	Script	Commit
Fig. 3	plot_traits_v1	fig3.py	a1b2c3d
Table 2	test_ids + outputs	eval.py	a1b2c3d

Rule of thumb

Deposit first, then submit. Reserve the DOI early. Ask a labmate to download the package and rebuild one figure.

What venues ask for (checked Nov 2026)

Venue	Hosting	Metadata	Watch out for
NeurIPS E&D track	Dataverse, Kaggle, HF, OpenML (or self-host)	Croissant core + RAI, required	data due with the paper, not later
CVPR / ECCV / WACV	public by camera-ready if claimed	not prescribed	double-blind: no names, ISU, grant IDs, GPS in EXIF
Scientific Data	approved repository before review	repository metadata + DOI	data + code availability sections
IMECE, ASABE, NAPPN	no rule	none	cite a DOI anyway; NSF/USDA expect sharing

Sources: NeurIPS 2026 E&D CFP + hosting guidelines; CVPR 2026 author guidelines; ECCV 2026 author practices; Scientific Data guide to authors.

The data availability statement

Open

Data are available in <Repository> at doi:<...> under CC BY 4.0. Code at <URL> (commit a1b2c3d), MIT license.

Partly restricted

Traits and metadata are open at doi:<...>. Genotype data are restricted under <agreement> and available on reasonable request.

Embargoed

Data will be released at doi:<...> on <date>, at most 12 months after publication. Metadata are public now.

Double-blind review

An anonymized copy is available to reviewers at <link>. A DOI will be provided at publication.

Your job before next group meeting

Pick your dataset (a–f).

Fill templates/study_card.yaml and data_dictionary.csv.

Arrange the folder in the lab layout (Folder Curator or CLI).

Score it: FAIR score + templates/readiness_score.md.

Bring two fields that did not fit your data.

Everything is in the repo

docs/02 formats
docs/03 examples
docs/04 paper packages
docs/06 our six datasets
templates/

In one line: collect about 20 facts once, on the day, and every format can be built from them.